

MINOR PROJECT 2

Project Overview

Zero Trust Architecture — Web Application

Project	Zero Trust Architecture — Web Application
Student	Himanshu Soni
Organisation	Navitech Solutions Pvt. Ltd.
Division	Cybersecurity
Standard	NIST SP 800-207
Year	2025

1. Introduction

This project implements a Zero Trust Architecture (ZTA) web application as a proof-of-concept for enterprise-grade security. It demonstrates how the principle of *"Never Trust, Always Verify"* can be applied in a real-world, containerised environment using widely available open-source technologies.

Developed by Himanshu Soni at Navitech Solutions Pvt. Ltd. (Cybersecurity Division), the system enforces strict role-based access control (RBAC) at every request boundary — no user, device, or service is trusted by default, even if it originates from inside the network perimeter.

2. Objectives

- Demonstrate the Zero Trust principle of continuous verification across all access requests.
- Implement role-based access control (RBAC) with least-privilege enforcement.
- Containerise the application using Docker for reproducible, isolated deployment.
- Block privilege escalation and unauthenticated access at the application layer.
- Provide a practical academic reference aligned with NIST SP 800-207.

3. Core ZTA Principles Applied

Principle	Implementation in this Project
Never Trust, Always Verify	Every page load re-validates the active session role client-side via localStorage
Least Privilege	Users access only Admin or Employee dashboard — nothing more
Micro-Segmentation	Admin and Employee dashboards are completely isolated HTML environments
Continuous Monitoring	All access attempts (success and failure) trigger alerts and logs
Minimal Attack Surface	Login portal exposes only username, password, and login — no role selector

4. Tech Stack

Layer	Technology	Purpose
Frontend	HTML5, JavaScript	UI, session role check, RBAC enforcement
Web Server	Nginx (latest)	Static file serving, containerised reverse proxy
Database	MySQL 8	User credential storage (production extension)
Containerisation	Docker	Isolated, reproducible deployment
Architecture Standard	NIST SP 800-207	Zero Trust reference framework

5. Project Structure

The project follows a minimal, security-first directory layout:

```
ZTA/  
  app/  
    login.html    - Single entry point for all users  
    admin.html    - Admin dashboard (admin-role only)  
    employee.html - Employee dashboard (employee-role only)  
  Dockerfile      - Nginx container build config
```

6. Application Roles

Role	Username	Dashboard Access	Capabilities
Admin	admin	admin.html	Manage Users, View Security Logs, Configure Policies
Employee	employee	employee.html	View Tasks, Update Profile, Submit Reports

7. Deployment

7.1 Docker Build & Run

```
docker build -t zta-webapp .  
docker run -d -p 8080:80 --name zta-container zta-webapp
```

The application is accessible at <http://localhost:8080> after the container starts. A docker-compose configuration supports simultaneous launch of the web app and MySQL database.

8. References

- [1] NIST SP 800-207 — Zero Trust Architecture (2020)
- [2] Kindervag, J. — Build Security Into Your Network's DNA — Forrester Research, 2010
- [3] Docker Documentation — Container Security & Networking
- [4] OWASP Authentication Cheat Sheet
- [5] Microsoft Zero Trust Architecture Framework
- [6] Gartner — SIEM & UEBA Best Practices for Enterprise Security Monitoring (2023)